STAC AI BOOTCAMP NYC



Evaluation and evidence attribution

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The decision this lecture supports

Notebook: `06_evaluation.ipynb`

Is this system ready for this task—and what should we fix first?

- A benchmark score estimates performance on its tested cases.
- A diagnostic test isolates a specific capability or failure.
- An evidence trace tells us which sources and operations support a claim.

Roadmap: from a claim to a defensible test

Stage	Question	Time
1. Define	What exactly must work?	10 min
2. Measure	Which observations establish success?	10 min
3. Intervene	What changes should matter?	15 min
4. Diagnose	Which dependency explains the failure?	15 min
5. Decide	What can we claim and release?	10 min

Five learning objectives

- Write a task contract with observable success and abstention conditions.
- Choose separate measures of correctness, support, stability and calibration.
- Design paired interventions with repeat controls and uncertainty estimates.
- Distinguish evidence dependencies from causal effects inside a model.
- Turn failures into a repair, retest and bounded release claim.

Start with an evaluation contract

Specify before testing	Why it is needed
Input and allowed evidence	Defines what the system may know.
Output and permitted abstention	Makes success observable, including missing information.
Target cases and exclusions	Defines where the result applies.
Errors, costs and release tolerance	Determines which failures matter operationally.

A benchmark is a sample of the task

- Cover ordinary cases, difficult cases and cases that cannot be answered.
- Include shortcuts that work on easy cases but fail on carefully matched cases.
- Keep independent test families separate from development examples.
- Report results by failure type as well as overall.

A labelled evaluation record

Field	Illustrative record
Case and family	E01; meeting-update family.
Input	The meeting moved from Tuesday to Friday. Which day now?
Reviewed reference	Friday.
Support label	Friday is supported; Tuesday is superseded.
Missing-source variant	Expected behavior: withhold the unsupported answer.

Keep case families separate from development. Do not train on the final evaluation fixtures.

Correctness, support and stability are different

Property	Question	Possible failure
Correctness	Does it match the reference?	Correct answer by lucky guessing.
Support	Do eligible sources justify it?	Plausible claim with irrelevant citation.
Stability	Does irrelevant variation change it?	Same facts, different answer order.
Calibration	Does stated probability match frequency?	Confident errors on unseen cases.

Choose the measurement for the claim

Target	Observable	What it does not establish
Exact extraction / arithmetic	Exact match; recomputed value	Whether the source is eligible.
Free-text support	Atomic claim + cited span + rubric	Whether the model relied on it.
Operational behavior	Executed tool and action trace	Whether the answer is true.
Language similarity	Overlap or embedding similarity	Entailment, factuality or authorization.

Evaluate the evaluator

- Write a rubric with correct, unsupported, contradicted and unanswerable examples.
- Have independent reviewers label a shared sample; inspect disagreement.
- If a model assists grading, validate it against reviewed labels and retain reasons.
- Use accountable checks and review for release decisions.

Question: a correct answer cites the right document

What have we established?

- A. The model must have used that document.
- B. Correctness and a citation are observed; dependence still needs testing.
- C. The answer will remain correct after the document changes.

Answer: B—test dependence separately

- Check that the cited passage actually supports the claim.
- Change or remove that evidence under a defined intervention.
- Compare the outcome with a matched control.
- Interpret the result only for the tested intervention.

Three kinds of behavioral test

Test	What should happen?	Simple example
Minimum functionality	A basic capability works.	Extract the stated date.
Invariance	Meaning-preserving edits preserve the result.	Change the order of independent records.
Directional expectation	Relevant changes move the result appropriately.	Replace positive evidence with a contradiction.

Control one difference at a time

Arm	Held fixed	Changed
Baseline repeats	Evidence, wording, order, settings	Execution only.
Presentation intervention	Evidence content and task	Framing or record order.
Evidence intervention	Task and presentation template	One relevant fact or missing source.
Negative control	Relevant evidence and task	An unrelated passage.

Measure changes, not just final labels

$$\text{flip} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{a_i \neq a_i'\}, \qquad \text{TV}(q_i, q_i') = \tfrac{1}{2}\sum_a |q_i(a) - q_i'(a)|.$$

- n: paired cases; i: case index; a_i, a_i': selected actions before/after the intervention; prime marks the changed condition.
- $\mathbf{1}\{\cdot\}$: 1 if true, 0 otherwise; $\neq$: differs from; Σ: sum; $|\cdot|$: absolute value.
- $q_i(a), q_i'(a)$: normalized probabilities of action a in the same action set; TV = total variation.
- Both measures range from 0 to 1, unitless. Zero flips need not mean zero probability movement; correctness must also be checked.

Candidate scoring requires a fixed boundary

$$\ell(a \mid x) = \sum_{t=1}^{T_a} \log p(a_t \mid x, a_{<t}), \qquad q(a \mid x) = \frac{\exp \ell(a \mid x)}{\Sigma_{b \in \mathcal{A}} \exp \ell(b \mid x)}.$$

- x: prompt plus answer prefix; a: candidate token string; t: token position; T_a: token count; a_t: next token; $a_{<t}$: earlier candidate tokens.
- p: model token probability; ℓ: summed natural log probability; log: natural logarithm; exp: its inverse.
- $\mathcal{A}$: candidate set; b: alternative in that set; q: normalized candidate score; $\mid$: conditional on; Σ: sum.
- All scores are unitless. Fix tokenization and termination for every candidate. These are not calibrated event probabilities; absent API alternatives are unknown.

Calibration needs outcomes and a separate split

$$\Pr(Y = 1 \mid \widehat{p} \approx 0.8) \approx 0.8$$

- Y: observed binary correctness indicator (1 = correct, 0 = incorrect).
- $\widehat{p}$: predicted probability; hat denotes an estimate; Pr: probability; $\mid$: conditional on; $\approx$: approximately.
- Read: among cases assigned about 80% confidence, roughly 80% should be correct. All quantities are unitless.
- Define the event; fit on reviewed calibration cases; assess on untouched representative cases and repeat after system changes.

Statistical evidence must match the test design

Comparison	Approach	Unit that must remain paired
Binary correctness, two systems	McNemar test; discordant cases	Same case under both systems.
Mean score difference	Paired bootstrap interval	Independent case or case family.
Many permutations per case	Summarize within case first	Do not count permutations as new cases.

A small p-value does not establish safe use

- Report effect size and an uncertainty interval.
- Set a practically acceptable sensitivity tolerance before looking at results.
- Failure to reject zero difference does not prove equivalence.
- Account for multiple tested prompts, metrics and model comparisons.

Class Exercise: specify a controlled evaluation

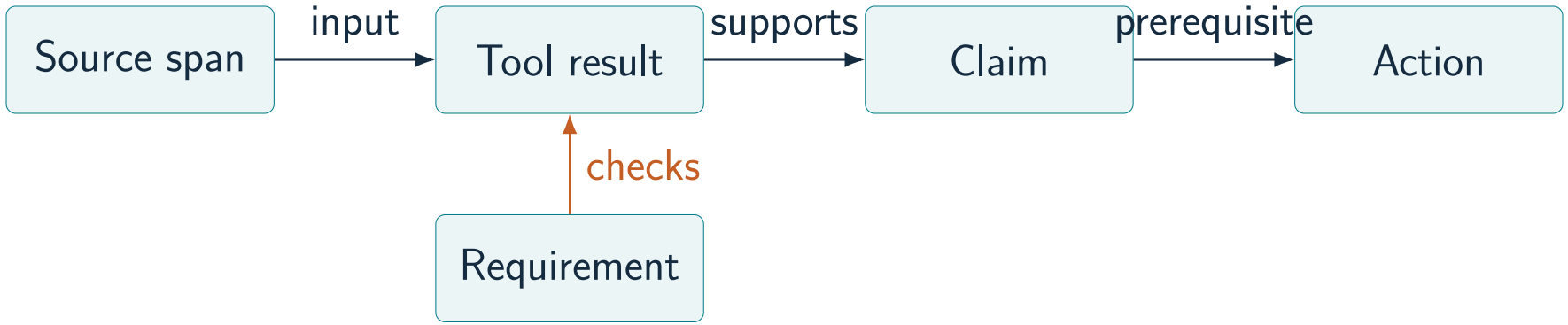
Step	Write down
1. Choose a claim	The answer should follow the supplied evidence.
2. Define three edits	Paraphrase; relevant contradiction; unrelated deletion.
3. Add a baseline	Repeat the identical prompt and exact evidence order.
4. Specify observations	Correctness, action flips and any available candidate scores.

Why inspect dependencies after scoring?

A failure rate tells us how often. A dependency trace helps identify what to repair.

Recorded object	Diagnostic question
Source span	Is this the fact the answer claims to use?
Tool input and output	Were the cited values actually passed to the calculator?
Derived claim	Which assumptions and prerequisites support it?
Proposed action	Which unresolved claims and permissions matter?

An evidence graph makes the repair path visible



- Record edge meanings and provenance; retain disputed links.
- A missing input marks downstream support unresolved. Recheck after repair.

Support, dependence and mechanism form a ladder

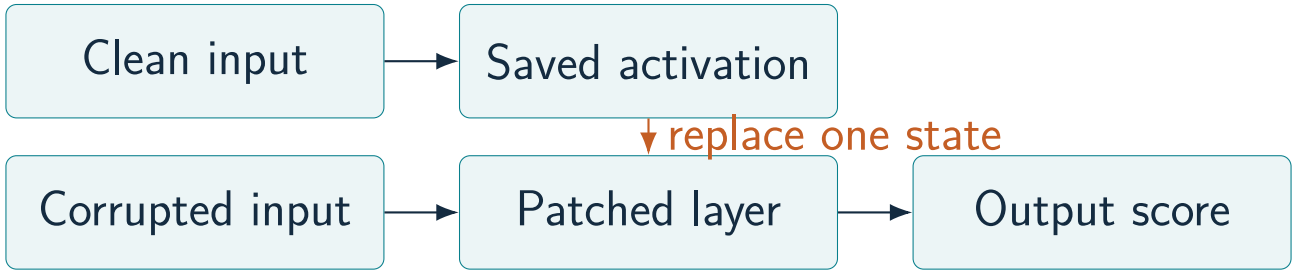
Claim	Evidence needed
The passage supports the answer	Reviewed entailment and eligibility.
Changing the passage changes the answer	Controlled input intervention.
A particular hidden state mediates the effect	Internal intervention with controls.
The repair improves deployment outcomes	Independent outcome evaluation.

Remove evidence; then retain only that evidence

$$\mathrm{Comp} = p_y(x) - p_y(x \setminus R), \qquad \mathrm{Suff} = p_y(x) - p_y(R).$$

- x: complete input; R: selected rationale evidence; $x \setminus R$: input with that evidence removed.
- y: fixed target answer; $p_y(z)$: its model probability under input z. Keep task instructions when retaining only R.
- Comp = comprehensiveness score drop on removal; Suff = sufficiency score drop using only the rationale. Differences are unitless probability points.
- Larger Comp suggests stronger removal effect; smaller Suff suggests retained support. Compare unrelated removals; redundancy and changed input distribution limit causal interpretation.

Patching asks a narrower internal question



$$\Delta = s(\text{corrupted run with patch}) - s(\text{corrupted run})$$

$s(\cdot)$ is a fixed output-score function; Δ is its patched-minus-original change. Use the same score and input; units follow the score. A positive change means the patch increased it.

Use the graph to select tests, then verify them

- Start from a failed claim and identify its source spans and tool inputs.
- Test a source-level repair before intervening inside the model.
- If needed, map the changed text to token positions for a controlled patch.
- Retest the repaired workflow on independent cases.

Question: no answer changes across reorderings

Which conclusion is warranted?

- A. The model is reliable in general.
- B. No selected-answer changes occurred in these tested cases.
- C. Its candidate probabilities must also be unchanged.

Answer: B—state the tested boundary

- Report the cases, permutations, repeats and output parser.
- Measure probability movement separately when complete scores are available.
- Check correctness: stable errors remain errors.
- More general reliability requires broader cases and an explicit tolerance.

Class Exercise: inspect saved framing responses

- Open `06_evaluation.ipynb`; run setup C01 and case C02.
- C03: compare neutral, bullish and skeptical responses and repeats.
- Change the case or model, rerun C02–C03; then C04 for all-case metrics.
- C05: remove actual earnings-per-share evidence from the checker.
- Inspect correctness separately from stability. C05 does not regenerate model answers.

Recorded pilot: low sensitivity can coexist with errors

Observed in the expanded framing pilot	Interpretation
1,200 assessments across two models	A controlled synthetic evaluation, not population prevalence.
Bullish framing did not increase unsupported buys	No evidence for that specific directional effect in this test.
Qwen mishandled some evidence gaps; Llama passed the benchmark	Correctness and framing sensitivity must be reported separately.

What BeliefLens adds to this evaluation loop

Activity	Concrete output
Organize evidence and executed steps	Traceable nodes, relations, source versions and tool bindings.
Apply declared requirements	Unresolved support, contradictions and affected claims.
Compare controlled runs	Case-level differences linked to evidence changes.
Repair and recheck	Recorded repair, review owner and remaining failures.

Finish with an evaluation card

Record	Decision it enables
Task, population and reference process	Know where the result applies.
Case families, controls and model revision	Reproduce and interpret the comparison.
Correctness, support, sensitivity and intervals	Assess behavior and uncertainty separately.
Known gaps, owner and retest trigger	Repair, limit or withhold deployment.

Five objectives: what can you now demonstrate?

Objective	Evidence from this session
Define success	Write the task contract and abstention condition.
Choose measures	Separate correctness, support, stability and calibration.
Control the comparison	Specify paired changes, repeats and uncertainty.
Diagnose responsibly	Distinguish graph dependencies from internal causal tests.
Decide what follows	Name a repair, retest and bounded release claim.

Reading route: lectures and primary methods

- Choi, Stanford CS224N Lecture 11 (2026) — benchmark design, metrics and evaluation protocols.
- Ribeiro et al., CheckList (ACL 2020) — minimum, invariance and directional tests.
- DeYoung et al., ERASER (ACL 2020) — rationale support and perturbation diagnostics.
- Dror et al., Statistical Significance in NLP (ACL 2018) — selecting paired significance tests.
- Guo et al., Calibration (ICML 2017); Meng et al., Causal Tracing / ROME (NeurIPS 2022) — confidence and internal interventions.